



IndianOil

JOURNEY RISK MANAGEMENT (JRM) STUDY

Salem Terminal TO K.S.K. & P.M.V. FUELS

Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable

recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

Compliance with the Road Transport Safety Policy (RTSP)

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

Emergency Preparedness and Response

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

Environmental Considerations

The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

Journey Risk Mitigation

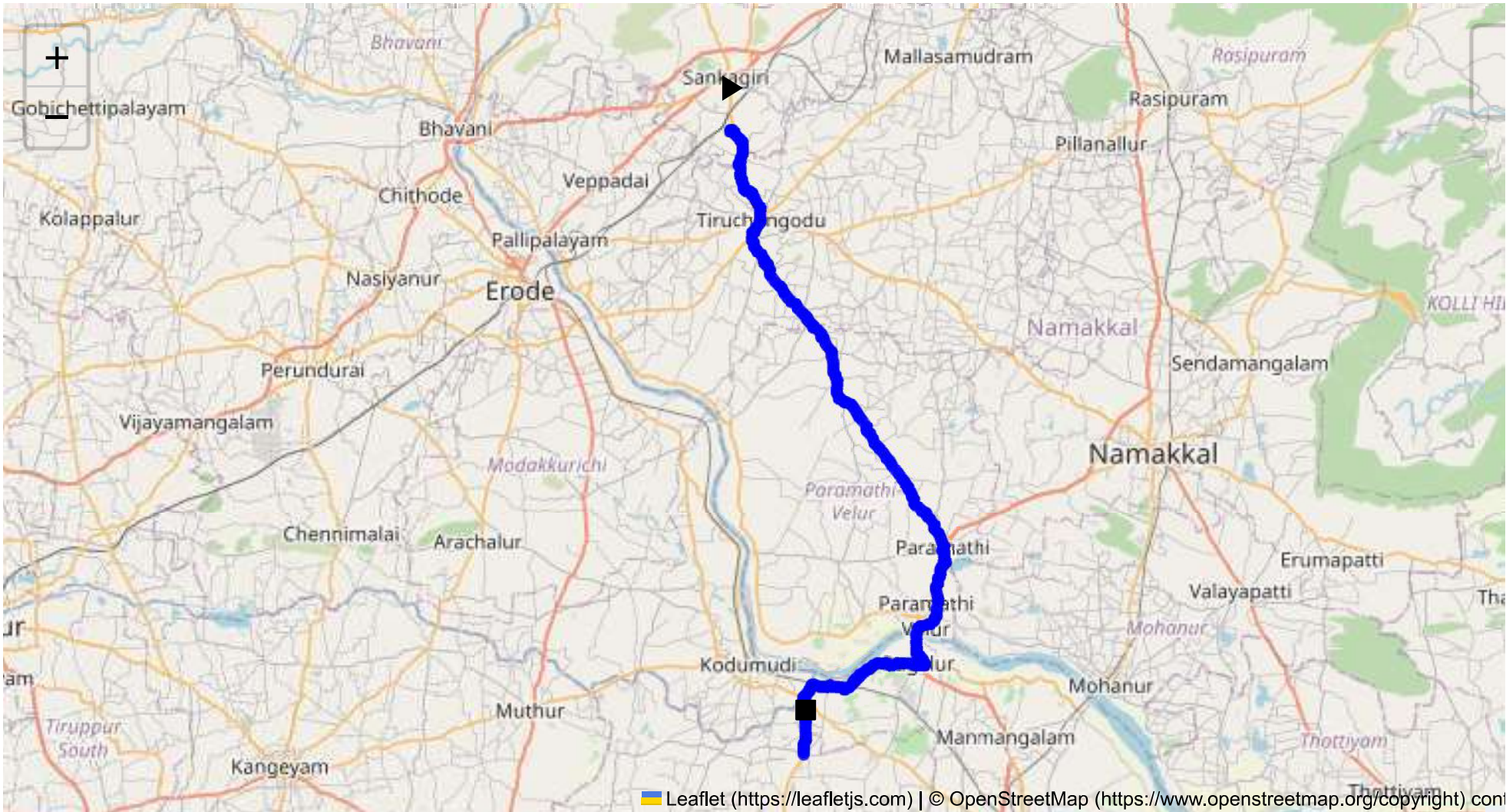
The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

Compliance with Government Directives

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.



Route Summary:
Total Distance: 62.86 km
Estimated Duration: 1.4 hours
Adjusted Duration (Heavy Vehicle): 1.7 hours
Start: (11.4381, 77.8734)
End: (11.014568, 77.924126)



Welcome to the Journey Risk Management Study

Route Safety Analysis: CVQF+23W, Sangagiri to 2W7F+RM Vallakolathupalayam, Tamil Nadu, India

1. Overview of the Route Map:

- The route spans approximately 62.86 kilometers, primarily utilizing major state highways. It travels through semi-urban and rural areas, potentially passing through small towns and villages.

The route is primarily straightforward but may include some curves and junctions typical of regional highways.

2. Typical Weather Conditions and Potential Weather-Related Hazards:

- The region experiences a tropical climate, with hot summers, moderate monsoons, and mild winters. During monsoon season (June to September), heavy rain can lead to slick roads, reduced visibility, and potential flooding in low-lying areas. Fog may occur occasionally in winter months.

3. Traffic Patterns:

- Peak hours are generally in the morning (7:00 AM - 9:00 AM) and evening (5:00 PM - 7:00 PM), coinciding with local business activities and school hours. Congestion may occur at town centers and junctions. Traffic is usually lighter in rural stretches but can become dense around market days or local events.

4. Assessment of Road Quality and Infrastructure:

- The primary roads on this route are generally well-maintained, but secondary roads might have potholes or uneven surfaces, especially after heavy rains. There may be narrow bridges and limited shoulders on rural roads.

5. Suggestions for Alternative Routes for Emergencies:

- In case of blockage or road work, use alternative local roads that parallel the main highways. Routes through nearby Nagar roads connecting local villages can offer bypasses. Consult local traffic advisories for real-time updates.

6. Summary of Local Regulations Affecting Hazardous Material Transport:

- Heavy vehicles transporting hazardous materials must adhere to strict guidelines regarding load securing, permissible hours of transport (typically restricting night travel), and designated routes. Permits are mandatory, and documentation must be readily available for inspection.

7. Overview of Historical Incidents:

- Records are sparse, but occasional reports indicate incidents involving overturned trucks or spillage due to sharp turns or sudden braking. Awareness initiatives have reduced such occurrences, emphasizing safe driving practices and maintenance checks.

8. Environmental Considerations and Sensitive Areas:

- Maintain reduced speeds near schools and hospitals which may be indicated by local signage. Some sections may pass by agricultural lands or wildlife reserves, requiring cautious driving to prevent accidents involving local fauna.

9. Analysis of Communication Coverage:

- Mobile networks are generally reliable along this route, with possible weaker signals in remote rural stretches. It is advisable to carry radio equipment as a backup for maintaining communication, especially in dead zones.

10. Estimated Emergency Response Times:

- Response times vary, with urban segments expecting quicker responses (approximately 20-30 minutes) due to proximity to towns, whereas rural segments might see delays (up to an hour) due to distance from medical and emergency services.

11. Overall Summary of Risk Assessment:

- Overall, this route is moderately safe for heavy vehicles carrying hazardous materials if standard precautions are observed. Key risks include weather conditions like rain, traffic during peak hours, and potential road wear. Regular weather checks, cautious driving, and awareness of local regulations can mitigate most risks. Ensuring communication and knowledge of alternative routes further enhance safety.

Regular training and updates based on the latest traffic regulations and infrastructural changes are recommended for drivers to maintain best safety practices.

Risk Assessment - Turns

	Risk Type	Risk Level	Coordinates	Speed Limit
0	Roundabout	High	11.38395, 77.89480	15 KM/Hr
1	Roundabout	High	11.07462, 78.00608	15 KM/Hr
2	Roundabout	High	11.07485, 78.00156	15 KM/Hr
3	Turn	Medium	11.43822, 77.87348	30 KM/Hr
4	Turn	High	11.43968, 77.87345	15 KM/Hr
5	Blind Spot	Blind Spot	11.44029, 77.87544	10 KM/Hr
6	Turn	High	11.37862, 77.89488	15 KM/Hr
7	Turn	High	11.37850, 77.89453	15 KM/Hr
8	Turn	Medium	11.16755, 78.01568	30 KM/Hr
9	Turn	High	11.14545, 78.02208	15 KM/Hr
10	Blind Spot	Blind Spot	11.14480, 78.02291	10 KM/Hr
11	Turn	High	11.07466, 78.00842	15 KM/Hr
12	Turn	High	11.06098, 77.93066	15 KM/Hr
13	Turn	High	11.05147, 77.92435	15 KM/Hr
14	Turn	Medium	11.05046, 77.92522	30 KM/Hr

Emergency Locations

	type	name	coordinates	speed_limit	risk_level
0	hospital	Tiruchengode, Goverment Hospital	11.3903328, 77.8920627	30 km/h	Medium
1	hospital	SPM Medical Centre,Tiruchengode	11.3881331, 77.8931963	30 km/h	Medium
5	hospital	T.C.A Hospital Tiruchengode	11.3791885, 77.8965774	30 km/h	Medium
6	clinic	Kongu Nursing Home	11.3783065, 77.8961134	30 km/h	Medium
7	hospital	Soorya Multispecialty Hospital	11.3786429, 77.8931912	30 km/h	Medium
8	hospital	Tiruchengode Government Hospital	11.37645, 77.89426	30 km/h	Medium
9	hospital	Krishna Hospital, Namakkal	11.3754811, 77.8931817	30 km/h	Medium
10	hospital	Tirukumaran Hospitals	11.3782118, 77.8914933	30 km/h	Medium
13	police	கரட்டுப்பாளையம் காவல் நிலையம்	11.357586, 77.8922539	30 km/h	Medium
14	hospital	Goverment Hospital, Nallur	11.2679698, 77.946863	30 km/h	Medium
16	hospital	Goverment Hospital	11.156647, 78.0213586	30 km/h	Medium
18	hospital	Government Hospital, Velauthapalayam	11.0729521, 78.0023123	30 km/h	Medium

Crowded Spots

	type	name	coordinates	speed_limit	risk_level
2	school	அரசு ஆண்கள் மேல்நிலைப் பள்ளி	11.3850439, 77.8948279	30 km/h	Medium
3	school	அரசு பெண்கள் மேல்நிலைப் பள்ளி	11.3844247, 77.8949053	30 km/h	Medium
4	marketplace	திருச்செங்கோடு தினசரி காய்கறி சந்தை	11.3833608, 77.8970145	30 km/h	Medium
11	school	KSR Educational institution	11.3772013, 77.8908807	30 km/h	Medium